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DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

PROJECT PHASE I (23AISE309)

Project Synopsis

on

" Neuro-Adaptive Closed-Loop System for Stroke Rehabilitation using Real-Time EEG and Computer Vision"

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Table of Contents

S.No		Details	Page No
1		Introduction	1-2
2		Literature Survey	3-12
	2.1	Existing Systems and their Limitations	8-9
	2.2	Prior Art Search	10
	2.3	Gaps Identified through Literature Survey	11-12
3		Problem Statement and Objectives	13-14
4		References	15-16

INTRODUCTION

1.INTRODUCTION

Stroke is one of the leading causes of long-term disability worldwide, often resulting in partial or complete loss of motor function. Damage to motor cortex regions disrupts voluntary movement control, significantly affecting a patient's quality of life. Effective rehabilitation is critical for promoting motor recovery and restoring functional independence.

Traditional rehabilitation approaches rely on repetitive physical exercises guided by therapists or digital platforms. While these methods improve muscle strength and coordination, they primarily evaluate progress based on observable movement rather than neural activation.

However, motor recovery fundamentally depends on neuroplasticity — the brain's ability to reorganize and form new neural connections. Without sufficient cortical engagement, rehabilitation outcomes may remain limited.

Recent advancements in Brain-Computer Interface (BCI), electroencephalography (EEG), and machine learning have enabled real-time monitoring and interpretation of brain activity. These technologies open new possibilities for designing intelligent, closed-loop rehabilitation systems that adapt therapy based on neural engagement rather than just physical performance.

This project proposes a Neuro-Adaptive Rehabilitation System for stroke patients that utilizes real-time EEG analysis and machine learning techniques to dynamically adjust exercise difficulty based on motor cortex activation. The aim is to enhance personalized therapy and promote effective neuroplastic recovery through brain-driven adaptive intervention.

LITERATURE SURVEY

2.LITERATURE SURVEY

To conduct the literature survey systematically, relevant research papers and journal articles related to EEG-based motor rehabilitation, brain–computer interface (BCI) systems, signal processing, and machine learning techniques were reviewed. Key focus areas included motor imagery analysis, sensorimotor rhythm extraction, real-time EEG processing, and adaptive rehabilitation frameworks. Reputable databases such as IEEE Xplore, Springer, ScienceDirect, and PubMed were used to gather recent and foundational studies. The reviewed works were analyzed to understand existing methodologies, limitations, and research gaps. This structured review helped establish a strong foundation for the proposed EEG-driven adaptive rehabilitation system.

S.No	Paper Title	Methodology	Outcome	Identification of Scope of Improvement	Year
1	Recommendations for Combining Brain-Computer Interface, Motor Imagery, and Virtual Reality in Upper Limb Stroke Rehabilitation: Qualitative Participatory Design Study.	Modeled Segmented Difficulty Scaling (SDS) logic for progressive cognitive loading. Utilized Qualitative Thematic Analysis to define task-logic variables.	Established 6 clinical design pillars, specifically advocating for Activity of Daily Living (ADL) based task modeling.	Does not incorporate real-time monitoring of cognitive engagement or cortical activation dynamics to adapt task difficulty based on evolving neural effort.	2025

2	A real time action scoring system for movement analysis and feedback in physical therapy using human pose estimation	Utilized MediaPipe Pose for 3D skeleton tracking. Calculated accuracy using Dynamic Time Warping (DTW) to measure the Euclidean distance between trajectories.	Demonstrated a high-precision real-time scoring engine (0–100) for physical therapy, enabling automated feedback.	Focuses solely on biomechanical accuracy and does not account for underlying neural intent or cognitive motor effort, limiting applicability for neurologically impaired patients.	2025
3	A hybrid EMG–EEG interface for robust intention detection and fatigue-adaptive control of an elbow rehabilitation robot	Implemented Wavelet Packet Transform (WPT) for feature extraction and a Fuzzy Logic Controller to merge EEG/EMG signals for adaptive robot control.	Significant increase in intention detection accuracy; proved that Sensor Fusion reduces error rates caused by muscle fatigue.	Relies heavily on peripheral muscle activation (EMG) and does not fully isolate central cortical effort, reducing applicability in patients with severe motor paralysis.	2025
4	Adaptive LDA Classifier Enhances Real-Time Control of an EEG Brain–Computer Interface for Decoding Imagined Syllables	Developed an Adaptive-LDA (A-LDA) that updates the weight vector w and bias b based on the Stochastic Gradient Descent (SGD) of incoming trial data.	Proved that Online Adaptation maintains higher classification accuracy (>85%) over long sessions despite neural signal drift.	While adaptive, the model does not evaluate dynamic cognitive–motor coupling or task-related cortical engagement during complex rehabilitative movements.	2024

5	Towards Transforming Neurorehabilitation: The Impact of Artificial Intelligence on Diagnosis and Treatment of Neurological Disorders	Analyzed CNN and RNN (LSTM) architectures for time-series EEG. Evaluated Random Forest and SVM for biomarker-based recovery prediction.	Validated that AI-driven models can identify patient-specific recovery patterns, moving toward Personalized Precision Medicine.	Emphasizes predictive modeling but does not integrate continuous neurophysiological state monitoring for real-time cognitive engagement tracking.	2024
6	Characterization of Event Related Desynchronization in Chronic Stroke Using Motor Imagery Based Brain Computer Interface for Upper Limb Rehabilitation	Computed Event-Related Desynchronization (ERD) via: $ERD\% = RA - R \times 100$. Used Welch's Periodogram for spectral density estimation.	Mapped spatial shifts in EEG activation, confirming Ipsilateral activation as a compensatory mechanism in chronic stroke.	Analyzes spectral desynchronization patterns but does not link ERD magnitude to concurrent behavioral output or cognitive effort modulation.	2024
7	Motor imagery brain-computer interface rehabilitation system enhances upper limb performance and improves brain activity in stroke patients: A clinical study	Applied Phase Lag Index (PLI) for functional connectivity analysis and correlated Mu-rhythm suppression with Fugl-Meyer (FM) clinical scores.	Proven that increased Neural Synchrony in the motor cortex is a direct bio-marker for significant motor function recovery.	Evaluates neural synchrony post-intervention without incorporating real-time assessment of cortical activation changes during active task execution.	2023

8	Gamified Neurorehabilitation Strategies for Post-stroke Motor Recovery: Challenges and Advantages	Integrated Behavioral Psychology Models with Reward-based Reinforcement Schedules within a game-engine framework.	Found that Gamification significantly increases patient compliance and "Time-on-Task," leading to better clinical outcomes.	Implements behavioral reinforcement mechanisms without integrating direct neurophysiological feedback reflecting cognitive load or motor intent strength.	2022
9	Toward an Adapted Neurofeedback for Post-stroke Motor Rehabilitation: State of the Art and Perspectives	Conducted a Systematic Review of Visual, Auditory, and Haptic feedback, analyzing Pooled Mean Differences via statistical forest plots.	Determined that Immersive Visual-VR feedback provides the highest "Sense of Agency" and triggers higher neural activation.	Primarily utilizes open-loop feedback paradigms and does not quantify the discrepancy between neural intent signals and executed motor output.	2022
10	Effects of brain-computer interface based training on post-stroke upper-limb rehabilitation: a meta-analysis	Evaluated Systematic Literature on Clinical BCI implementation, focusing on the transition from lab-based to Home-Based Systems.	Identified Hybrid Systems (combining sensors) and Cloud-based Processing as the future of scalable stroke rehab.	Highlights hybrid and scalable systems but does not propose a unified framework for simultaneous cognitive signal interpretation and movement-based validation using low-cost hardware.	2025

Table 2: Literature Survey

2.1 Existing Systems and Their Limitations

From the literature survey, various existing EEG-based rehabilitation and brain–computer interface (BCI) systems focus on motor imagery classification, assistive control, robotic rehabilitation, and performance-based therapy monitoring. These systems can be categorized based on their primary focus areas and identified limitations:

System Type	Focus Area	Existing Limitations
Conventional Physiotherapy	Physical motor rehabilitation through therapist-guided exercises	Uses generalized exercise protocols; depends on therapist observation; lacks objective neural measurement; no real-time personalization based on brain activity.
Robotic-Assisted Rehabilitation Systems	Assisted limb movement using robotic exoskeletons or motorized devices	Focuses on mechanical movement execution; does not assess cortical engagement; high cost; limited adaptability to individual neural recovery.
Virtual Reality (VR)-Based Rehabilitation	Interactive motor training using immersive virtual environments	Evaluates task completion and movement accuracy; lacks real-time neural feedback; adaptation is performance-based not brain-based
EMG-Based Rehabilitation Systems	Muscle activity monitoring during rehabilitation	Measures peripheral muscle activation rather than central neural activity; cannot determine motor cortex engagement; limited insight into neuroplastic recovery
Brain-Computer Interface (BCI) Motor Imagery Systems	Detection of motor intent using EEG signals	Primarily designed for assistive communication or prosthetic control; limited integration with adaptive rehabilitation mechanisms

Fixed Protocol Digital Rehab Applications	Pre-programmed digital exercise platforms	Follow predefined therapy levels; lack patient-specific neural adaptation; do not incorporate real-time EEG driven feedback
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Table 2.1: Existing Systems and Their Limitations

2.2 Prior Art Search

The literature survey highlights prior research efforts in neuroplasticity and rehabilitation frameworks. Key developments include:

- **Commercial BCI Systems (e.g., Recoverix, IpsiHand):**
 - Current State: Successfully uses Motor Imagery (MI) to trigger muscle stimulation (FES) or robotic orthotics using the 8–30 Hz Mu-rhythm.
 - The Gap: These are "Static Threshold" systems. They lack real-time difficulty adjustment based on neural fatigue and do not verify physical movement via external cameras (Computer Vision).
- **Adaptive Biofeedback Patents (e.g., US9827411B2, EP3458148A1):**
 - Current State: Protects the logic of "Intent-Triggered-Action" and adjusts visual feedback based on the Signal-to-Noise Ratio (SNR) of brainwaves.
 - The Gap: Adaptation is limited to the EEG signal alone. There is no Multi-Modal Data Fusion to cross-reference mental intent with physical joint trajectories (Pose Estimation).
- **AI-Driven Pose Estimation (e.g., MediaPipe-based Frameworks):**
 - Current State: Uses standard webcams to track joint angles and calculate Range of Motion (ROM) with high accuracy.
 - The Gap: These systems are "Brain-Blind." For a paralyzed patient, the system records a "zero" score because it cannot detect the Internal Mental Effort when physical execution fails.

Innovation & Novelty Statement

- The "Integration Gap": Existing prior art treats EEG (Brain) and Computer Vision (Body) as separate silos.
- The Solution: This project introduces a Neuro-Motor Congruence Score.
- The Novelty: By mathematically aligning the ERD Equation (Neural Intent) with the DTW Accuracy Score (Physical Performance), the system creates a personalized, adaptive loop that rewards Mental Effort even when physical movement is limited.

2.3 Gaps Identified Through Literature Survey

1. Lack of Multi-Modal Data Synchronization (The "Signal Silo" Gap):
 - The Gap: Current rehabilitation systems typically function in silos—either tracking physical movement via cameras or monitoring brain activity via EEG. There is no real-time mathematical alignment between Neural Intent and Physical Execution.
 - The Impact: A patient can "cheat" a camera-based exercise using compensatory movements (e.g., using the shoulder to lift a paralyzed arm), and the system will incorrectly reward the movement without the corresponding neural activation.
2. Absence of Neural-Driven "Partial Credit" (The "Binary Feedback" Gap):
 - The Gap: Most AI-based scoring systems are performance-oriented. If a hemiplegic patient attempts to move but physically fails, standard pose-estimation systems record a "zero" score.
 - The Impact: This leads to "learned helplessness." Existing systems fail to detect and reward the Internal Mental Effort (Event-Related Desynchronization) when external execution is absent, which is critical for early-stage neuroplasticity.
3. Static Thresholding and Lack of Adaptive Scaling (The "Fatigue" Gap):
 - The Gap: Standard BCI and CV systems use fixed calibration baselines. They do not account for the non-stationary nature of neural signals or the physical fatigue that occurs during a 30-minute session.
 - The Impact: As a patient tires, their "Mental Effort" signal weakens. A static system will stop recognizing their intent, leading to frustration and session abandonment rather than adapting the difficulty to the patient's current state.
4. High Technical Complexity and Cost Barriers (The "Accessibility" Gap):
 - The Gap: Clinical-grade neuro-rehabilitation requires expensive 64-channel "wet" EEG caps and robotic exoskeletons that are difficult to

set up without professional help.

- The Impact: This restricts effective therapy to high-end clinics. There is a clear lack of a "Zero-Contact" home-based solution that combines standard webcams with consumer-grade dry EEG headbands.

5. Non-Congruent Visual Feedback (The "Agency" Gap):

- The Gap: In many gamified systems, the virtual avatar moves based on pre-set timers or simple triggers, rather than being strictly contingent on the patient's simultaneous neural and physical "try."
- The Impact: For the brain to "rewire" itself, the visual feedback must be temporally congruent with the intent. Existing systems often lack this tight synchronization, reducing the effectiveness of the therapy.

PROBLEM STATEMENT AND OBJECTIVES

4. PROBLEM STATEMENT AND OBJECTIVES

Problem Statement

Stroke causes motor impairment due to damage in motor cortex regions. Existing rehabilitation systems use generalized exercise protocols and evaluate progress based only on physical movement, without measuring neural engagement. Since motor recovery depends on active cortical participation and neuroplasticity, the absence of real-time brain monitoring limits therapy personalization and effectiveness. Therefore, there is a need for a closed-loop rehabilitation system that utilizes real-time EEG analysis and machine learning to adapt exercise difficulty based on neural activation.

Objectives

The proposed project aims to develop a functional, real-time neuro-adaptive rehabilitation prototype by achieving the following milestones:

- Establish a Multi-Modal Data Acquisition Pipeline: To synchronize live EEG streams (via LSL protocol) with real-time body tracking data using a standard webcam and MediaPipe Pose Estimation.
- Implement Real-Time Neural Feature Extraction: To isolate Event-Related Desynchronization (ERD) in the Mu (8Hz - 13Hz) and Beta (13Hz-30Hz) bands as reliable indicators of a patient's "Mental Try."
- Develop a Subject-Specific Intent Classifier: To design a lightweight Machine Learning model (such as a Linear Discriminant Analysis - LDA) that can be calibrated in 60 seconds to recognize the specific neural patterns of the user.
- Construct a Neuro-Motor Congruence Scoring Engine: To create a mathematical "Match Score" that verifies if the patient's physical arm movement (tracked by the camera) is happening at the exact same time as their neural intent (tracked by the EEG).
- Design a Performance Accommodation Mechanism (PAM): To develop a logic-based loop that automatically makes the virtual exercise "easier" (requires less movement) if the system detects the patient is trying hard mentally but struggling physically.

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